

SEESAW DUAL REDUCTIVE PAIRS

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Introduction:

In this paper I want to discuss, in an informal style, a certain structure connected with the θ -correspondence for dual reductive pairs [5], or, in more classical terminology, the theory of theta-functions. This structure, which I call a seesaw dual reductive pair or seesaw pair, gives rise to a certain family of identities between inner products of automorphic forms on different groups. Identities of precisely this type occur in disguised form in many places in the literature; but since they frequently result from complicated manipulations, they often appear rather mysterious. On the other hand, once the seesaw structure has been recognized, such identities arise very formally and there remains only the problem of interpreting or computing both sides. Of course it is often the case that previous complications are now concentrated in this step. Nonetheless, the seesaw structure makes the existence of certain identities clear from the outset and thus sheds light on their nature.

Section 1 contains the definition of seesaw pairs and a description of the basic identity to which they give rise. Section 2 contains a list of some typical see-saw pairs. The first aim of this paper then is to illustrate how the basic identity occurs in with some familiar situations - Section 3. On the other hand, the occurrence of a seesaw pair often suggests new identities or generalizations and analogues of old ones. A few speculations of this sort are contained in Section 3.

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1. Seesaw pairs and the basic identity.

Let $W, <, >$ be a non-degenerate symplectic vector space over a field k of characteristic zero (for convenience), and let $Sp(W)$ be the corresponding symplectic group. Recall that a dual reductive pair (G, G') in $Sp(W)$ is a pair of reductive subgroups of $Sp(W)$ which are mutual centralizers, i.e.

$$Z_{Sp(W)}(G) = G'$$

and

$$Z_{Sp(W)}(G') = G.$$

In certain situations it will be convenient to slightly weaken the conditions on (G, G') as follows:

Definition 1.1. A weakly dual reductive pair (G, G') in $Sp(W)$ is a pair of reductive subgroups such that there exists a finite field extension K/k , a symplectic vector space $W_0, <, >_0$ over K and a dual reductive pair (H, H') in $Sp(W_0)$ such that:

$$(i) \quad W \simeq RW_0$$

and

$$<, > = \text{tr}_{K/k} <, >_0$$

where $R = R_{K/k}$ denotes restriction of scalars from K to k ; and

(ii) the natural homomorphism

$$RSp(W_0) \rightarrow Sp(RW_0) \cong Sp(W)$$

induces isomorphisms

$$RH \xrightarrow{\sim} G$$

and

$$RH' \xrightarrow{\sim} G'.$$

For convenience we will usually shorten "dual reductive pair" and "weakly dual reductive pair" to "dual pair" and "weakly dual pair" respectively.

Now suppose that k is a number field and let

$$\widetilde{Sp(W_{\mathbb{A}})} \rightarrow Sp(W_{\mathbb{A}}) \quad (1.1)$$

be the non-trivial 2-fold central extension of the adelization of $Sp(W_{\mathbb{A}})$. Fix a non-trivial additive character ψ of $k_{\mathbb{A}}/k$ and let $\omega = \omega_{\psi}$ denote the oscillator representation of $\widetilde{Sp(W_{\mathbb{A}})}$ determined by ψ . Let S denote the space of smooth vectors in some (e.g. Schrödinger) model of ω and let Θ be the Θ -distribution on S . For example, if

$$W = W' + W'' \quad (1.2)$$

is a complete polarization of W , i.e. decomposition into maximal isotropic subspaces, then we may take

$$S = S(W'_{\mathbb{A}}) \quad (1.3)$$

to be the space of Schwartz-Bruhat functions on $W'_{\mathbb{A}}$ and, for $\varphi \in S$,

$$\Theta(\varphi) = \sum_{x \in W'_k} \varphi(x). \quad (1.4)$$

The covering (1.1) has a unique splitting over $Sp(W_k)$, and, identifying $Sp(W_k)$ with its image under this splitting,

$$\Theta(\omega(g)\varphi) = \Theta(\varphi) \quad \forall g \in Sp(W_k), \varphi \in S. \quad (1.5)$$

Now if (G, G') is a weakly dual pair in $Sp(W)$, let $\widetilde{G(\mathbb{A})}$ and $\widetilde{G'(\mathbb{A})}$ denote the full inverse images of $G(\mathbb{A})$ and $G'(\mathbb{A})$ in $\widetilde{Sp(W_{\mathbb{A}})}$. For $\varphi \in S, g \in \widetilde{G(\mathbb{A})}$ and $g' \in \widetilde{G'(\mathbb{A})}$ let

$$\theta_{\varphi}(g, g') = \Theta(\omega(g)\omega(g')\varphi). \quad (1.6)$$

If f is a cusp form on $\widetilde{G(\mathbb{A})}$ and if φ is finite for a suitable compact subgroup of $G'(\mathbb{A})$, then

$$\theta_{\varphi}(f)(g') = \int_{G(k) \backslash G(\mathbb{A})} f(g) \theta_{\varphi}(g, g') dg \quad (1.7)$$

is an automorphic form on $\widetilde{G'(\mathbb{A})}$. For more details the reader should consult [5] and [6], or for a more classical discussion [7,13,14,15,22, 27 etc]. Similarly, if f' is a cusp form on $\widetilde{G'(\mathbb{A})}$, and if φ satisfies a certain finiteness condition, then:

$$\overline{\theta}_{\varphi}(f)(g') = \int_{G'(k) \backslash \widetilde{G'(\mathbb{A})}} f'(g) \overline{\theta_{\varphi}(g, g')} dg \quad (1.8)$$

is an automorphic form on $\widetilde{G(\mathbb{A})}$.

If a pair of cusp form (f, f') is given, then one clearly has the following adjointness formula:

$$\langle \theta_{\varphi}(f), f' \rangle_{G'} = \langle f, \overline{\theta}_{\varphi}(f') \rangle_G \quad (1.9)$$

where, for example, for automorphic forms f_1, f_2 on $\widetilde{G(\mathbb{A})}$ with, say f_1 a cusp forms, and with the same central characters:

$$\langle f_1, f_2 \rangle_G = \int_{\widetilde{Z(\mathbb{A})} G(k) \backslash G(\mathbb{A})} f_1(g) \overline{f_2(g)} dg \quad (1.10)$$

with respect to Tamagawa measure dg on $\widetilde{G(\mathbb{A})}$, where $\widetilde{Z}$ is the center of $\widetilde{G(\mathbb{A})}$.

The resulting relations between automorphic forms f and $\theta_{\varphi}(f)$ or f' and $\overline{\theta}_{\varphi}(f')$ will be called the θ -correspondence. It can be expressed in terms of automorphic representations, as well, but this refinement is not necessary for our informal discussion - see [5,6,17].

If (G, G') is a weakly dual pair arising by base change from a dual pair (H, H') in $Sp(W_0)$ as in Definition 1.1, then there are isomorphisms:

$$G(\mathbb{A}) \simeq RH(\mathbb{A}) \simeq H(\mathbb{A}_K) \quad (1.11)$$

and

$$G'(\mathbb{A}) \simeq RH'(\mathbb{A}) \simeq H(\mathbb{A}_K) \quad (1.12)$$

where $\mathbb{A}_K$ denotes the adèle ring of K and $\mathbb{A}$ that of k . It is then easy to check the following:

Lemma 1.2: (i) The inverse image of $RSp(W_0)(\mathbb{A}) \simeq Sp(W_0, \mathbb{A}_K)$ in $\widetilde{Sp(W_{\mathbb{A}})}$ is isomorphic to the covering $\widetilde{Sp(W_0, \mathbb{A}_K)}$ given by (1.1).
(ii) If

$$\psi_K = \psi_{\text{otr}_{K/k}}$$

then, as representations of $\widetilde{Sp(W_0, \mathbb{A}_K)}$,

$$\iota^*(\omega_{\psi}) \simeq \omega_{\psi_K}.$$

In fact, if

$$W_0 = W'_0 + W''_0$$

is a complete polarization of $W_0, <, >_0$ as a K vector space and if

$$W \cong RW'_0 + RW''_0$$

is the corresponding complete polarization of $W, <, >$ over k , then the natural isomorphism

$$W'_{\mathbb{A}} = (RW'_0)_{\mathbb{A}} \simeq W'_{0, \mathbb{A}_K}$$

induces an isomorphism

$$S(W'_{\mathbb{A}}) \simeq S(W'_{0, \mathbb{A}_K})$$

which intertwines $\iota^*(\omega_{\psi})$ and ω_{ψ_K} .

(iii) Under the isomorphisms (1.11) and (1.12) the theta correspondence associated to ψ and the weakly dual pair (G, G') in $Sp(W)$ coin-

cides with that associated to ψ_K and the dual pair (H, H') in $Sp(W_0)$. □

Thus all of the nice properties (correspondence of Hecke eigenforms, Howe duality conjecture, etc.) of the θ -correspondence for dual pairs actually hold for the θ -correspondence associated to weakly dual pairs.

Definition 1.3. A seesaw (weakly) dual pair in $Sp(W)$ is a pair (G, H') , (H, G') of (weakly) dual pairs in $Sp(W)$ such that

$$G \supset H \quad \text{and} \quad G' \supset H'.$$

The picture which motivates this terminology^{*} is the following:

$$\begin{array}{ccc} G & & G' \\ | & \diagdown & / \\ & H & H' \\ | & / & \diagdown \\ H & & H' \end{array} \quad (1.13)$$

where the sloping lines connect members of a dual pair and the vertical lines indicate inclusions.

The basic identity associated to such a pair is an immediate generalization of the adjointness formula (1.9). If f (resp. f') is a cusp form on $\widetilde{H(\mathbb{A})}$ (resp. $\widetilde{H'(\mathbb{A})}$), then for $\varphi \in S$:

$$\boxed{\langle \theta_{\varphi}(f), f' \rangle_{H'} = \langle f, \bar{\theta}_{\varphi}(f') \rangle_H}. \quad (1.14)$$

Here we note that the forms $\theta_{\varphi}(f)$ and $\bar{\theta}_{\varphi}(f')$ have been restricted to $\widetilde{H'(\mathbb{A})}$ and $\widetilde{H(\mathbb{A})}$ respectively. It should also be noted that the two theta-kernels involved here:

$$\theta_{\varphi}(g, h') \quad (1.15)$$

^{*}seesaw: a plank balanced on a support at the middle, used by children at play, one sitting at either end and causing his end to rise and fall alternately with the other. Webster's New Twentieth Century Dictionary of the English Language, 2nd ed., New York, 1979.

and

$$\theta_{\varphi}(h, g'') \quad (1.16)$$

have a common specialization

$$\theta_{\varphi}(h, h'),$$

but that no function " $\theta_{\varphi}(g, g')$ " exists. It is perhaps this fact which gives the completely formal identity (1.14) its 'mysterious' quality—particularly if the functions (1.15) and (1.16) are written out in classical notation.

Finally it should be remarked that the restriction that f and f' be cusp forms is unfortunate and ought, ultimately, to be removed. For example, the origin of the whole subject is the Weil-Siegel formula which says that, under certain restrictions on (G, G') (cf. [24, 25])

$$\theta_{\varphi}(\mathbb{1}) = E'_{\varphi} \quad (1.17)$$

where $\mathbb{1}$ is the constant function on $\widetilde{G(\mathbb{A})}$ and E'_{φ} is a certain Eisenstein series on $\widetilde{G'(\mathbb{A})}$. It would be worthwhile to find, even conjectural, analogues of (1.17) in which $\mathbb{1}$ is replaced by an arbitrary Eisenstein series or residue of such a series. Some results in this direction are contained in [18].

2. Examples of seesaw pairs.

Before turning to the literature it will be best to have in hand some typical examples of seesaw pairs. It would, of course, be nice to have a complete classification, but such a classification will probably be rather complicated.

First suppose that $W, <, >$ (resp. $V, (,)$) is a k vector space with a non-degenerate alternating (resp. symmetric) bilinear form; let

$$\widetilde{W} = V \otimes_k W \quad (2.1)$$

and

$$\langle \cdot, \cdot \rangle^{\sim} = (\cdot, \cdot) \otimes \langle \cdot, \cdot \rangle \quad (2.2)$$

so that, in the standard way, $(O(V), Sp(W))$ forms a dual pair in $Sp(\tilde{W})$. Then, for any decomposition

$$W = W_1 + W_2 \quad (2.3)$$

of W into symplectic subspaces which are perpendicular with respect to $\langle \cdot, \cdot \rangle$, we obtain a seesaw pair:

$$\begin{array}{ccc} Sp(W) & & O(V) \times O(V) \\ | & \searrow & | \\ Sp(W_1) \times Sp(W_2) & & O(V) \end{array} \quad (2.4)$$

in $Sp(\tilde{W})$ where the notation is as in (1.13). Similarly, if

$$V = V_1 + V_2 \quad (2.5)$$

is any orthogonal decomposition of V with respect to $(\cdot, \cdot)$, then we obtain a seesaw pair

$$\begin{array}{ccc} Sp(W) \times Sp(W) & & O(V) \\ | & \searrow & | \\ Sp(W) & & O(V_1) \times O(V_2) \end{array} \quad (2.6)$$

in $Sp(\tilde{W})$. These seesaw pairs might be called "direct sum" type.

Next suppose that K/k is a quadratic extension with Galois group generated by σ , and let $R = R_{K/k}$ denote restriction of scalars as in Section 1. Let $W^*, \langle \cdot, \cdot \rangle^*$ (resp. $V^*, (\cdot, \cdot)^*$) be a σ -skew Hermitian left (resp. σ -Hermitian right) vector space over K , let

$$\tilde{W} = V^* \otimes_K W^* \quad (2.7)$$

and let

$$\langle , \rangle^{\sim} = \text{tr}_{K/k}(\langle , \rangle \otimes \langle , \rangle^{\sigma}). \quad (2.8)$$

Then $(U(V^*), U(W^*))$ is a dual pair in $\text{Sp}(\widetilde{W})$ (viewed as algebraic groups over k). Now suppose that $V^*, (\langle , \rangle)^*$ is the σ -Hermitian extension of a symmetric k -bilinear space $V, (\langle , \rangle)$, i.e.

$$V^* \simeq V \otimes_k K \quad (2.9)$$

and

$$(v \otimes \alpha, v' \otimes \beta)^* = \alpha^{\sigma}(v, v')\beta.$$

Then there is a natural isomorphism of symplectic vector spaces over k :

$$\widetilde{W} = V^* \otimes_K W^* \simeq V \otimes_k RW^* \quad (2.10)$$

where RW^* is given the symplectic form $\text{tr}_{K/k}\langle , \rangle^*$, and this isomorphism gives rise to a seesaw dual pair

$$\begin{array}{ccc} U(V^*) & & \text{Sp}(RW^*) \\ | & \searrow & | \\ O(V) & & U(W^*) \end{array} \quad (2.11)$$

in $\text{Sp}(\widetilde{W})$. Similarly, if $W^*, \langle , \rangle^*$ is the σ -skew Hermitian extension of a symplectic vector space $W, \langle , \rangle$ over k , then there is a natural isomorphism

$$\widetilde{W} = V^* \otimes_K W^* \simeq RV^* \otimes_k W \quad (2.12)$$

where RV^* is given the bilinear form $\text{tr}_{K/k}(\langle , \rangle)^*$. Again this isomorphism gives rise to a seesaw dual pair:

$$\begin{array}{ccc} O(RV^*) & & U(W^*) \\ | & \searrow & | \\ U(V^*) & & \text{Sp}(W) \end{array} . \quad (2.13)$$

Of course the same construction could be made with a more general algebra with involution in place of (K, σ) .

Next suppose that k/k_0 is an arbitrary finite extension and let $W, <, >$ and $V, (,)$ be as in the first example. Suppose that there exists $V_0, (,)_0$ over k_0 such that

$$V \simeq V_0 \otimes_{k_0} k \quad (2.14)$$

and

$$(,) = k\text{-bilinear extension of } (,)_0.$$

Then there is a natural isomorphism of symplectic vector spaces over k_0 :

$$R\tilde{W} = R(V \otimes_k W) \simeq V_0 \otimes_{k_0} RW \quad (2.15)$$

where $R = R_{k/k_0}$ denotes restriction of scalars and the symplectic forms are

$$\text{tr}_{k/k_0}(<, >) \simeq (,)_0 \otimes \text{tr}_{k/k_0}(<, >). \quad (2.16)$$

Then there is a seesaw (weakly) dual pair

$$\begin{array}{ccc} RO(V) & & Sp(RW) \\ | & \searrow & | \\ O(V_0) & & RSp(W) \end{array} \quad (2.17)$$

in $Sp(R\tilde{W})$ where $(O(V_0), Sp(RW))$ is a dual pair in the usual sense and $(RO(V), RSp(W))$ is the weakly dual pair in $Sp(R\tilde{W})$ which arises from the dual pair $(O(V), Sp(W))$ in $Sp(\tilde{W})$ via restriction of scalars - See Definition 1.1 and Lemma 1.2. Similarly, since there always exists a symplectic vector space $W_0, <, >_0$ over k_0 whose k -bilinear extension is isomorphic to $W, <, >$, there exists, for each choice of $W_0, <, >_0$, and

$$W \simeq W_0 \otimes_{k_0} k \quad (2.18)$$

such that

$$\langle , \rangle = k\text{-bilinear extension of } \langle , \rangle_0,$$

a seesaw (weakly) dual pair

$$\begin{array}{ccc} O(RV) & & RSp(W) \\ | & \searrow & | \\ RO(V) & & Sp(W_0) \end{array} \quad (2.19)$$

in $Sp(\tilde{W})$.

Finally we consider examples which involve dual pairs of type (GL, GL) .

First recall that if U_1 and U_2 are vector spaces over k and if U_1^* and U_2^* are their duals, then

$$\tilde{W} = U_1 \otimes U_2^* + U_1^* \otimes U_2 \quad (2.20)$$

becomes a symplectic vector space with respect to the skew symmetric extension of the natural pairing

$$U_1 \otimes U_2^* \times U_1^* \otimes U_2 \rightarrow k. \quad (2.21)$$

This gives rise to a dual pair $(GL(U_1), GL(U_2))$ in $Sp(\tilde{W})$.

Now if $V, (,)$ and $W, \langle , \rangle$ are as in example 1 and if a polarization (1.2) of $W, \langle , \rangle$ is chosen, then there are natural isomorphisms

$$\tilde{W} = V \otimes W \simeq V \otimes W' + V \otimes W'' \simeq V \otimes (W'')^* + V^* \otimes W'' \quad (2.22)$$

where the isomorphisms $W' \simeq (W'')^*$ and $V \simeq V^*$ are determined by $\langle , \rangle$ and $(,)$ respectively. The isomorphism (2.22) yields a seesaw dual pair:

$$\begin{array}{ccc}
 \text{GL}(V) & & \text{Sp}(W) \\
 | & \searrow & | \\
 \text{O}(V) & & \text{GL}(W''')
 \end{array}
 \quad (2.23)$$

Next suppose that U_1, U_2 and U are k vector spaces with dual U_1^*, U_2^* and U^* . For the moment let

$$\tilde{U}_1 = U_1 \otimes U$$

and

$$\tilde{U}_2 = U^* \otimes U_2.$$

Then there is a natural isomorphism of symplectic vector spaces

$$\tilde{W} = \tilde{U}_1 \otimes U_2^* + \tilde{U}_1^* \otimes U_2 \simeq U_1 \otimes \tilde{U}_2^* + U_1^* \otimes \tilde{U}_2 \quad (2.24)$$

which gives rise to a seesaw pair

$$\begin{array}{ccc}
 \text{GL}(U_1 \otimes U) & & \text{GL}(U^* \otimes U_2) \\
 | & \searrow & | \\
 \text{GL}(U_1) & & \text{GL}(U_2)
 \end{array}
 \quad (2.25)$$

in $\text{Sp}(\tilde{W})$.

Finally there is an analogue of (2.25) which involves base change. Let k/k_0 be a finite extension and let U_1^0 and U_2^0 be vector spaces over k_0 with k_0 -linear duals U_1^{0*} and U_2^{0*} . Let

$$U_i = U_i^0 \otimes_{k_0} k \quad (2.26)$$

and let U_i^* be the k -linear dual of U_i . Then there are natural isomorphisms

$$U_i^* \simeq (U_i^{0*}) \otimes_{k_0} k \quad (2.27)$$

and

$$R(U_1^*) \simeq (RU_1)^* \quad (2.28)$$

where $R = R_{k/k_0}$ denotes restriction of scalars and this last isomorphism is obtained by composing with tr_{k/k_0} . Setting

$$\tilde{W} = U_1 \otimes_k U_2^* + U_1^* \otimes_k U_2 \quad (2.29)$$

there are natural isomorphisms:

$$\begin{aligned} R(W) &= R(U_1 \otimes_k U_2^*) + R(U_1^* \otimes_k U_2) \\ &\simeq U_1^0 \otimes_{k_0} R(U_2^*) + U^{*0} \otimes_k R(U_2) \\ &\simeq U_1^0 \otimes_{k_0} R(U_2)^* + U^{*0} \otimes_k R(U_2), \end{aligned} \quad (2.30)$$

which preserve the symplectic form $\text{tr}_{k/k_0}(<, >^{\sim})$. (2.30) and its analogue with the roles of U_1 and U_2 reverse give rise to a seesaw pair

$$\begin{array}{ccc} \text{GL}(RU_1) & & \text{GL}(RU_2) \\ | & \searrow & | \\ \text{RGL}(U_1) & \text{---} & \text{RGL}(U_2) \\ | & \nearrow & | \\ \text{GL}(U_1^0) & & \text{GL}(U_2^0) \end{array} \quad (2.31)$$

with three positions.

The list of examples given here is, no doubt, far from complete. Other possibilities are suggested by the results of Satake [19].

3. Examples of the basic identity.

In this section I will describe the role played by a seesaw pair and the corresponding basic identity in a number of familiar examples. I describe these examples in a very informal way, often glossing over fundamental difficulties such as convergence, or neglecting fundamental results. My aim is simply to highlight the part played by the seesaw pair.

Example 1:

Consider the seesaw pair

$$\begin{array}{ccc}
 & \text{Sp}(1) & \text{GL}(V) \\
 & \downarrow & \downarrow \\
 \xi_s & \text{GL}(1) & \text{O}(V)
 \end{array}
 \quad
 \begin{array}{l}
 \bar{\theta}'_\phi(\xi_s) = E_\phi(s) \\
 \chi
 \end{array}
 \quad (3.1)$$

where we assume that $\dim V$ is even. First suppose that $V, (,)$ is anisotropic at infinity so that an automorphic form χ on $\text{O}(V)$ is essentially a function on generalized 'classes' of lattices in a genus, and $\theta_\phi(\chi)$ is a classical theta function. In general $\theta_\phi(\chi)$ is not a cusp form, but, deleting a constant term if necessary (this deletion is indicated by '), we may take the Mellin transform $\langle \theta_\phi(\chi)', \xi_s \rangle$ where $\xi_s, s \in \mathbb{C}$, is a family of characters of $\text{GL}(1)$. The seesaw pair identity then gives

$$\langle \theta_\phi(\chi)', \xi_s \rangle = \langle \chi, \bar{\theta}'_\phi(\xi_s) \rangle, \quad (3.2)$$

and it is easy to check that $\bar{\theta}'_\phi(\xi_s) = E_\phi(s)$ is a very simple Eisenstein series for $\text{GL}(V)$.

For example, if $V, (,)$ is the norm form for an imaginary quadratic field $K/\mathbb{Q}$ and if χ is trivial, then the Mellin transform of $\theta_\phi(\chi)'$ is, up to a constant and Γ -factors, the zeta function $\zeta_K(s)$ of K , and (3.2) amounts to the usual expression of $\zeta_K(s)$ as a sum, over a finite set of CM points, of the values of the standard Epstein type zeta function.

Similarly, if $V, (,)$ is the norm form for a real quadratic field $k/\mathbb{Q}$ and if χ is trivial, then (3.2) reduces to Hecke's [4] expression for $\zeta_k(s)$ as a sum of integrals of the standard Epstein zeta function over a finite set of hyperbolic arcs associated to the ideal classes of k .

These elementary examples suggest that it would be interesting to consider other $V, (,)$ which are isotropic, at least at some infinite places. As a semi-classical example, consider:

$$\begin{array}{ccccc}
 \theta_{\varphi}(f) & & \text{Sp}(1) & & \text{GL}(m+2) & & \overline{\theta}'_{\varphi}(\xi_s) = E_{\varphi}(s) \\
 & & \downarrow & \nearrow & \downarrow & & \\
 \xi_s & & \text{GL}(1) & & \text{O}(m,2) & & f
 \end{array} \quad (3.3)$$

where m is even, and where the map $f \mapsto \theta_{\varphi}(f)$ was studied by Oda [14]. The identity

$$\langle \theta_{\varphi}(f)', \xi_s \rangle = \langle f, E_{\varphi}(s) \rangle \quad (3.4)$$

then relates the Mellin transform of $\theta_{\varphi}(f)$ to the integral of f against the restriction of an Eisenstein series from $\text{GL}(m+2)$. In this situation, a study of the residues or values of $E_{\varphi}(s)$ might lead to some interesting results.

Finally it should be noted that a similar seesaw pair:

$$\begin{array}{ccc}
 \text{Sp}(n) & & \text{GL}(V) \\
 \downarrow & \nearrow & \downarrow \\
 \text{GL}(n) & & \text{O}(V)
 \end{array} \quad (3.5)$$

with $V(,)$ anisotropic at infinity occurs in Maass [11, Chapt. 17].

Example 2:

For the Shimura-Shintani-Niwa-Waldspurger [20,22,13,23] correspondence the basic dual pair is $(\widetilde{\text{Sp}(1)}, \text{O}(1,2))$ where $\text{O}(1,2)$ is the orthogonal group associated to the discriminant form on the vector space of binary quadratic forms. Shimura considered the map from cusp forms on $\widetilde{\text{Sp}(1)}$ to forms on $\text{O}(1,2)$ via Weil's converse theorem while Niwa utilized the θ -correspondence. The two approaches are related by the following see-saw pair:

$$\begin{array}{ccccc}
 & & \text{Sp}(1) \times \widetilde{\text{Sp}(1)} & & \text{O}(1,2) \\
 & & \downarrow & \nearrow & \downarrow \\
 f & & \text{Sp}(1) & & \text{O}(1,1) \times \text{O}(1) & & \chi(s)
 \end{array} \quad (3.6)$$

where $\text{O}(1,1)$ is taken to be split, $\chi(s)$ is a character of $\text{O}(1,1)$ depending on $s \in \mathbb{C}$, ξ is a character of $\text{O}(1)$ and f is a cusp form of half-integral weight. Then the basic identity says that

$$\langle \theta_{\varphi}(f), \chi(s)\xi \rangle = \langle f, \overline{\theta}_{\varphi}(\chi(x)\xi) \rangle \quad (3.7)$$

where the left hand side is, in effect, the Mellin transform of the θ -lift $\theta_\varphi(f)$ of f . On the other hand, if $\varphi = \varphi_1 \otimes \varphi_2$, then

$$\overline{\theta}_\varphi(\chi(s)\xi) = \overline{\theta}_{\varphi_1}(\chi(s))\overline{\theta}_{\varphi_2}(\xi) \quad (3.8)$$

and we may interpret these functions as follows:

$$\overline{\theta}_{\varphi_1}(\chi(s)) = E(s) \quad (3.9)$$

is an Eisenstein series on $Sp(1)$ and $\overline{\theta}_{\varphi_2}(\xi)$ is a theta-function of the type which Shimura calls $h_\psi(z)$ [20]. Thus, by (3.7), the Mellin transform of $\theta_\varphi(f)$ has a Rankin type integral representation precisely as in Shimura [20]. Details may be found in Niwa [13]. Very complete information about the θ -correspondence in this case has been obtained by Waldspurger [23]. See example 4 for further remarks.

Example 3:

The theta correspondence for the dual pair $(Sp(1), O(2,2))$ gives, essentially, the base change lifting for a (real) quadratic extension $k/\mathbb{Q}$ [1,7,14,15,26,27]. Again the original approach of Doi-Naganuma [3] and Naganuma [12] is related to the θ -correspondence by a seesaw pair:

$$\begin{array}{ccccc} & Sp(1) \times Sp(1) & & O(2,2) & \theta_\varphi(f) \\ & \downarrow & \swarrow & \downarrow & \\ f & Sp(1) & & O(1,1) \times O(1,1) & \xi_s, \chi \end{array} \quad (3.10)$$

where the first $O(1,1)$ is split and the second corresponds to the norm form of $k/\mathbb{Q}$. Then, as in example 1, if $\varphi = \varphi_1 \otimes \varphi_2$, we find that $\overline{\theta}'_{\varphi_1}(\xi_s)$ is an Eisenstein series and $\overline{\theta}_{\varphi_2}(\chi)$ is a Maass wave form associated to $k/\mathbb{Q}$. The seesaw identity becomes:

$$\langle \theta_\varphi(f), \xi_s \rangle = \langle f, \overline{\theta}'_{\varphi_1}(\xi_s) \overline{\theta}_{\varphi_2}(\chi) \rangle \quad (3.11)$$

which says that the Mellin transform of $\theta_\phi(f)$ is given by the Rankin type integral studied in [3] and [12]. Some details may be found, for example, in [7].

On the other hand we can consider the seesaw pair

$$\begin{array}{ccccc}
 & \widetilde{\text{Sp}(1)} \times \widetilde{\text{Sp}(1)} & & \text{O}(2,2) & \\
 & \downarrow & \searrow & \downarrow & \\
 f & \text{Sp}(1) & & \text{O}(2,1) \times \text{O}(1) & E(s)
 \end{array} \quad (3.12)$$

where $E(s)$ is an Eisenstein series on the split $\text{O}(2,1)$ and $\mathbb{1}$ is the trivial character of $\text{O}(1)$. Formally we have a seesaw identity

$$\langle \theta_\phi(f), E(s) \rangle = \langle f, \bar{\theta}_\phi(E(s)) \rangle. \quad (3.13)$$

The left hand side is then a certain Dirichlet series with Euler product, associated to $\theta_\phi(f)$, essentially the quotient of the symmetric square series of $\theta_\phi(f)$ by that of f , [1][27]. Of course, the quantity $\bar{\theta}_\phi(E(s))$ on the right hand side is not well defined, but since f is holomorphic we have

$$\theta_\phi(f) = \langle f, \theta_\phi \rangle = \langle f, \Omega \rangle \quad (3.14)$$

where Ω is Zagier's kernel function [27, p. 152]. Therefore, in the 'identity' (3.13) we may replace the unfortunate $\bar{\theta}_\phi(E(s))|_{\text{Sp}(1)}$ by $\langle \Omega, E(s) \rangle$. This function has been computed by Zagier [27, Thm 4] (his $\phi_{s,D}(z)$), provided s lies in a certain vertical strip. The modified identity (3.13) may then be evaluated at integers in that strip and interesting arithmetic information, for example about

$$\frac{\langle \theta_\phi(f), \theta_\phi(f) \rangle}{\langle f, f \rangle^2} \quad (3.15)$$

comes out [27, p. 159, Cor. 1 and 2]. One might ask, on the basis of (3.13), if Zagier's $\phi_{s,D}$ is related to an Eisenstein series on $\text{Sp}(1)$?

Example 4:

Another example of the seesaw identity occurs in recent work of

Rallis [16,18]. He considers the problem of computing the inner product of θ -lifts on $O(V)$. The seesaw pair in question is the following:

$$\begin{array}{ccc}
 & \text{Sp}(W) & O(V) \times O(V) \\
 & \downarrow & \downarrow \\
 f_1, f_2 & \text{Sp}(W_1) \times \text{Sp}(W_2) & O(V)
 \end{array}
 \quad \theta_{\varphi_1}(f_1), \theta_{\varphi_2}(f_2) \quad (3.16)$$

where f_1 and f_2 are cusp forms and one of $\theta_{\varphi_1}(f_1)$ and $\theta_{\varphi_2}(f_2)$ is assumed to be cuspidal. Then we have:

$$\langle \theta_{\varphi_1}(f_1), \theta_{\varphi_2}(f_2) \rangle = \langle \theta_{\varphi_1}(f_1) \overline{\theta_{\varphi_2}(f_2)}, 1 \rangle = \langle \theta_{\varphi}(f_1 \otimes \bar{f}_2), 1 \rangle \quad (3.17)$$

for a suitable φ , (we are suppressing an involution which relates ω and $\bar{\omega}$). Now the Weil-Siegel formula implies that the last term is

$$\langle f_1 \otimes \bar{f}_2, E \rangle = \langle f_1 \otimes \bar{f}_2, E_{\varphi}(s) \rangle \Big|_{s=0} \quad (3.18)$$

where $E_{\varphi} = E_{\varphi}(s) \Big|_{s=0} = \overline{\theta_{\varphi}}(1)$ is a special value of an Eisenstein series attached to the maximal parabolic P of $\text{Sp}(W)$ with abelian unipotent radical. Using the structure of the double coset space $P \backslash \text{Sp}(W) / \text{Sp}(W_1) \times \text{Sp}(W_2)$, Rallis concludes that $\theta_{\varphi_1}(f_1)$ and $\theta_{\varphi_2}(f_2)$ are orthogonal unless $W_1 = W_2$ and in that case only the term of E_{φ} associated to the 'open' orbit in $P \backslash \text{Sp}(W)$ contributes. Ultimately, in the case when $\dim W_1 = 2$ and for $O(V)$ of rank 1, Rallis shows that $\langle f \otimes \bar{f}, E_{\varphi}(s) \rangle \Big|_{s=0}$ is a product of a special value of a Langlands L -function associated to f and a factor determined by local information about Whittaker models [16,18]. This work generalizes results of Waldspurger [23] and involves, as well, an extension of the Weil-Siegel formula beyond the previously known range! The seesaw identity plays a small yet important role.

Example 5:

The seesaw pairs described in (2.19) and (2.11) occur in Shimura [21]. He considers a totally real field F and an inner product space

$V, (,)$ over F of signature

$$\underbrace{(m, 2), \dots, (m, 2)}_r, \underbrace{(m+2, 0), \dots, (m+2, 0)}_{n-r} \quad (3.19)$$

where $|F:\mathbb{Q}| = n, r > 0$ and m is even. He assumes that F has a subfield E with $|E:\mathbb{Q}| = r$ and such that the first r embeddings of F into $\mathbb{R}$ have distinct restrictions to E . Let $R = R_{F/E}$ denote the restriction of scalars and let $W_0, <, >_0$ be a two-dimensional symplectic vector space over E . Then, letting $W = W_0 \otimes_E F$ with the F -bilinear extension of $<, >_0$, (2.19) becomes

$$\begin{array}{ccc} & \text{Sp}(W) & \text{O}(RV) \\ & \downarrow & \downarrow \\ f & \text{Sp}(W_0) & \text{O}(V) \end{array} \quad \theta_\phi(f) \quad (3.20)$$

Shimura's assumptions insure that $RV, \text{tr}_{F/E} (,)$ has signature:

$$(qm+2(q-1), \dots, qm+2(q-1)), \quad (3.21)$$

where $n = rq$, and that the embedding of Hermitian symmetric spaces induced by the inclusion $\text{O}(V) \rightarrow \text{O}(RV)$ is holomorphic. If f is a holomorphic cusp form on $\text{Sp}(W_0)$ and if $E(s)$ is a certain Eisenstein series of weight zero, the θ -lift $\theta_\phi(fE(s))$ is holomorphic at $s = 0$ and the value there is $\theta_\phi(f)$, a holomorphic function on $\text{O}(RV)$, and hence, by restriction, on $\text{O}(V)$ [21, section 13].

To study the arithmeticity of the lift $\theta_\phi(f)$, Shimura considers its value at a certain type of CM-point. In the simplest case these arise as follows: Let K be a CM field containing F with $|K:F| = m+2$ and assume that there exists a Hermitian space $V^*, (,)^*$ over K with $\dim_K V^* = 1$ and such that there is an isomorphism:

$$V \simeq R_{K/F} V^*$$

with

$$(,) \simeq \text{tr}_{K/F} (,)^*.$$

Then the seesaw pair (2.11) becomes:

$$\begin{array}{ccc}
 & U(W^*) & O(V) \\
 & | & | \\
 f & \text{Sp}(W) & U(V^*)
 \end{array}
 \quad (3.22)$$

where $W^* = W \otimes_F K$ with the skew-Hermitian extension of $\langle, \rangle$. Let χ be a grossencharacter of $U(V^*)$ and consider the 'value' of $\theta_\varphi(f)$:

$$\langle \theta_\varphi(f), \chi \rangle = \langle f, \bar{\theta}_\varphi(\chi) \rangle. \quad (3.23)$$

But now $\bar{\theta}_\varphi(\chi)$ is essentially a Hecke type binary theta-series on $U(W^*)$, and the rationality properties of the left hand side of (3.23) can be deduced from results about the rationality of Petersson inner products like that on the right hand side. Note that if we include the Eisenstein series $E(s)$, (3.23) becomes:

$$\langle \theta_\varphi(fE(s)), \chi \rangle = \langle fE(s), \bar{\theta}_\varphi(\chi) \rangle \quad (3.24)$$

and the right hand side of this identity is a Rankin type integral for a Dirichlet series of the type considered by Shimura [21, (9.2)]. The seesaw identity (3.24) thus exhibits the connection between rationality properties of special values of this series and the arithmeticity of $\theta_\varphi(f)$. Of course we have only described special cases of the results of [21], but it seems likely that the more general cases can be expressed in a similar way.

Example 6:

In [10] the seesaw pair identity is used to prove a rationality result about the periods of certain integrals. The seesaw pair in question is:

$$\begin{array}{ccc}
 & \text{Sp}(n) & U(n, 1) \\
 & | & | \\
 f & U(n) & O(n, 1)
 \end{array}
 \quad \omega = \theta_\varphi(f) \quad (3.25)$$

constructed as in (2.11) with respect to an imaginary quadratic extension $K/\mathbb{Q}$. Here the group $U(n)$ is anisotropic at infinity and, for a suitable choice of f and ϕ , the θ -lift $\theta_\phi(f) = \omega$ is a holomorphic $(n,0)$ -form on an arithmetic quotient of the complex n -ball. The period of ω over the real n -cycle determined by $O(n,1)$ can then be interpreted, via the seesaw identity, as

$$\langle \theta_\phi(f), \mathbb{1} \rangle = \langle f, \overline{\theta}_\phi(\mathbb{1}) \rangle. \quad (3.26)$$

On the other hand, the form $\overline{\theta}_\phi(\mathbb{1})$ was computed explicitly in [8] and [9] and is, for suitable normalizations, arithmetic. The rationality result then follows via Shimura's theorem about the special values (the right side of (3.26)) of such automorphic forms. Previous examples suggest that these values should be related to values of L -functions for $U(n,1)$.

Example 7:

This final example arises in connection with recent work of Bump and Goldfeld [2]. Let $K/\mathbb{Q}$ be a totally real cubic extension and let U_1 be a one dimensional K vector space. Also let U_2^0 be a two dimensional $\mathbb{Q}$ -vector space and let $U_2 = U_2^0 \otimes_{\mathbb{Q}} K$. Then (2.30) yields a seesaw pair:

$$\begin{array}{ccc} & GL_2(K) & GL_3(\mathbb{Q}) \\ & \downarrow & \downarrow \\ E(\phi) & GL_2(\mathbb{Q}) & K^\times \end{array} \quad (3.27)$$

For convenience let $G = RGL_2, G' = GL_3, H = GL_2$ and $H' = RGL_1$, where $R = R_{K/\mathbb{Q}}$ denotes restriction of scalars, so that (3.27) coincides with (1.13). Let Z be the center of H and let Z' be the image of $GL_1/\mathbb{Q}$ in H' . Also let

$$X = M_{2,3}(\mathbb{Q}) \quad (3.28)$$

and

$$Y = M_{2,1}(K) \quad (3.29)$$

so that, fixing a Q -basis for K , we obtain an isomorphism

$$Y \xrightarrow{\sim} X \quad (3.30)$$

compatible with the natural action of $H \times H'$ on both sides. Finally let X' (resp. Y') denote the subset of matrices of maximal rank in X (resp. Y). Then taking $\phi \in S(X_{\mathbb{A}})$, an Eisenstein series $E(\phi)$ on $H(\mathbb{A})$, and a character ω of $H'(\mathbb{A})$ we may form θ -lifts

$$\theta_{\phi, X'}(E(\phi)) \quad (3.31)$$

and

$$\theta_{\phi, Y'}(\omega) \quad (3.32)$$

where the subscript X' (resp Y') indicates that $\theta_{\phi, X'}(h, g')$ (resp. $\theta_{\phi, Y'}(g, h')$) is formed by summing over X'_Q (resp. Y'_K). An easy computation shows that

$$\theta_{\phi, X'}(E(\phi)) = E(\psi) \quad (3.33)$$

where $E(\psi)$ is an Eisenstein series formed from a function ψ on $G'(\mathbb{A})$ determined by ϕ and ω and which is left $B'(\mathbb{A})$ -equivariant. Assuming that the central character of $E(\psi)$ is the inverse of the restriction of ω to $Z'(\mathbb{A})$ we may consider

$$\langle \theta_{\phi, X'}(E(\phi)), \omega \rangle = \langle E(\psi), \omega \rangle = \langle E(\phi), \theta_{\phi, X'}(\omega) \rangle \quad (3.34)$$

where the 2nd equality follows from the seesaw identity. Here inner products are taken over $H''(\mathbb{A})/Z'(\mathbb{A})$ and $H(\mathbb{A})/Z(\mathbb{A})$ respectively. Now the series $\theta_{\phi, X'}(\omega)$ is a component of the restriction of $\theta_{\phi, Y'}(\omega)$ to $H(\mathbb{A})$ and $\theta_{\phi, Y'}(\omega)$ is an Eisenstein series on $G(\mathbb{A})$. Thus the second equality of (3.34) is the identity of Bump and Goldfeld. It is clear that many generalizations of this identity can be obtained from the seesaw construction. Details will appear elsewhere.

All of the examples above involve the global θ -correspondence.

Of course the seesaw machinery may also be applied to the local θ -correspondence, and in this case the identity (1.14) becomes a kind of reciprocity law. Results in the real case may be found in [28].

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